Structure, Bonding, Resonance & Naming: Practice

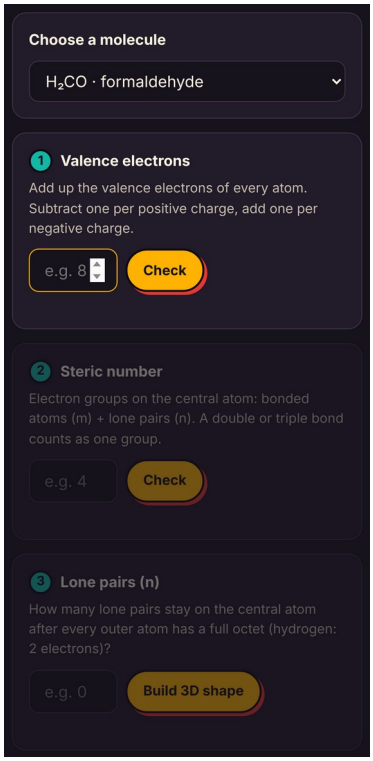
Name _____ Date _____

This packet goes with the Structure, Bonding, Resonance & Naming guide. Part 1 walks you through the VSEPR Builder. Part 2 is practice. Work each problem on paper first.

Part 1. Walk through the VSEPR Builder

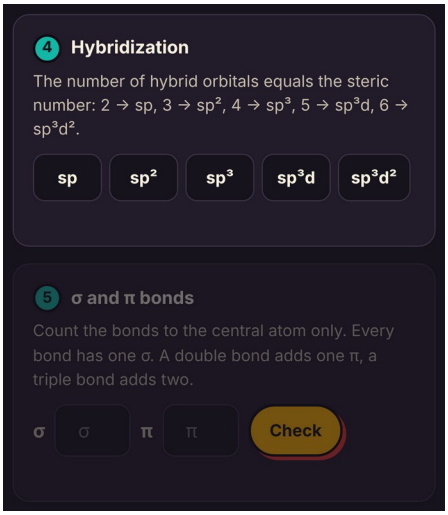
The VSEPR Builder (chemustad.org/vsepr.html) turns the counting rules from the guide into a 3D model you can rotate. Open it on a laptop, tablet, or phone. The numbered cards in the Builder match the steps here. Work each step on paper first, then check it in the Builder.

Step 1. Build formaldehyde. Under Choose a molecule, pick H_2CO · formaldehyde. Fill in card 1 (valence electrons), card 2 (steric number) and card 3 (lone pairs), then press Build 3D shape.



(a) Valence electrons ____ Steric number of C ____ Shape _____ Bond angle ____

Step 2. Hybridization. In card 4, pick the hybridization of the C.



(b) Hybridization of C _____

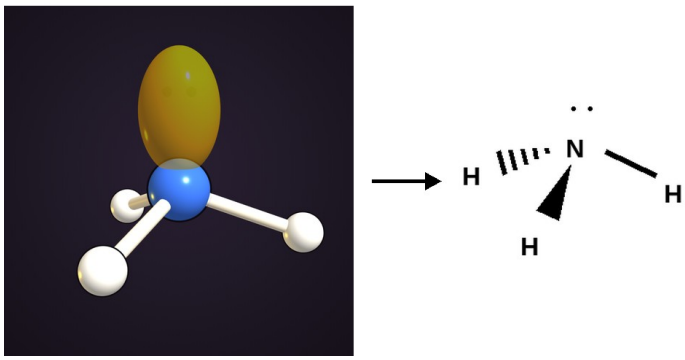
Why does the C=O double bond count as only one group? _____

Step 3. σ and π bonds. In card 5, enter the number of σ and π bonds on the C.

(c) σ ____ π ____ Which bond holds the π bond? _____

Step 4. Lone pairs change the shape. Pick CH_4, then NH_3, then H_2O. All three have a steric number of 4, so all three central atoms are sp^3. Rotate each model and look at where the lone pairs sit.

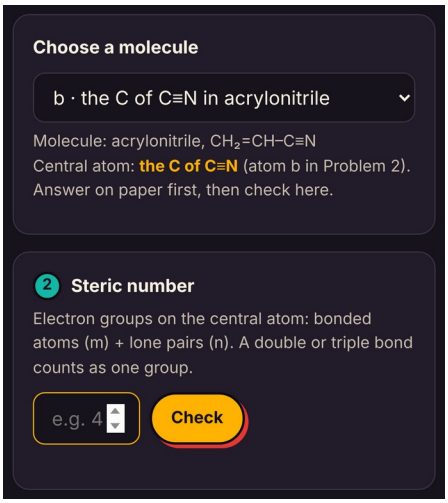
VSEPR Builder, NH3 **On paper**



(d) Shape and bond angle: CH_4 _____ NH_3 _____ H_2O _____

(e) Why does the bond angle get smaller from CH_4 to H_2O? _____

Step 5. Atoms inside bigger molecules. At the bottom of the molecule list is a section called Structure & Bonding practice, Problem 2. These are the labeled atoms a–k from Problem 2. For an atom inside a bigger molecule, card 1 is skipped and you start at the steric number. Do Problem 2 on paper first, then check each atom here.



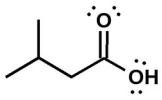
(f) Which atom does the Builder treat as an exception? _____ Where is its lone pair? _____

Part 2. Practice problems

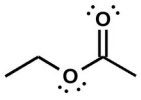
Work through these in order, with the guide next to you. A model kit helps for the shape questions, and so does the VSEPR Builder at chemustad.org.

1. Convert the structures. (a)–(d) Draw a bond-line structure for each condensed formula. (e)–(h) Write a condensed formula and the molecular formula for each bond-line structure.

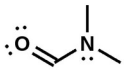
(a) $CH_3CH_2CH(CH_3)CH_2Br$ (b) $(CH_3)_3COH$ (c) $CH_3COCH_2CH_3$ (d) $CH_3CH{=}CHCOOH$



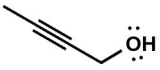
(e)



(f)



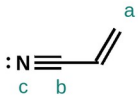
(g)



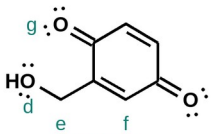
(h)

(e) _____ (f) _____ (g) _____ (h) _____

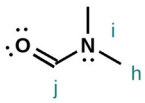
2. Hybridization and shape. Give the hybridization of each labeled atom (a–k). For each labeled carbon, also give the bond angle around it. When you are done, check each atom in the VSEPR Builder (Step 5 of the walkthrough).



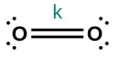
acrylonitrile



HMQ A



DMF



CO_2

a _____ b _____ c _____ d _____ e _____ f _____

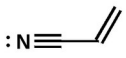
g _____ h _____ i _____ j _____ k _____

Hint for i: look at the resonance structures of DMF in the Resonance section of the guide.

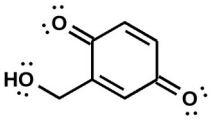
3. Count the bonds. Give the molecular formula and the number of σ bonds and π bonds in each molecule. Remember the hidden C–H bonds.



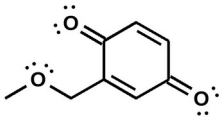
(a)
CO_2



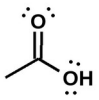
(b)
acrylonitrile



(c)
HMQ A



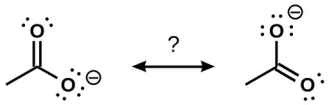
(d)
MMQ



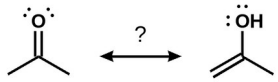
(e)
acetic acid

	(a)	(b)	(c)	(d)	(e)
formula					
σ bonds					
π bonds					

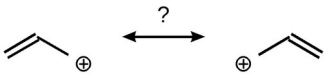
4. Are these resonance structures? For each pair, write yes or no. If no, say what rule is broken.



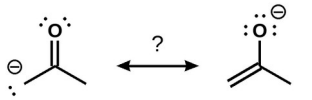
(a) _____



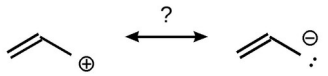
(b) _____



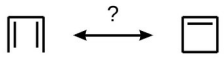
(c) _____



(d) _____

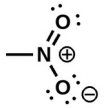


(e) _____

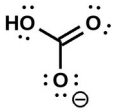


(f) _____

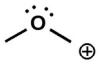
5. Draw the resonance structures. Draw all the reasonable resonance structures for each species, with curved arrows. If they are not equal, circle the major contributor.



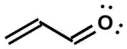
(a)



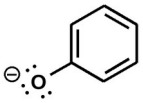
(b)



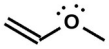
(c)



(d)

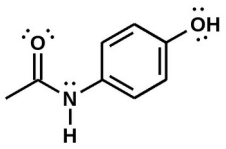


(e)

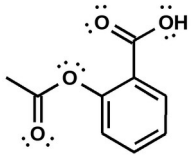


(f)

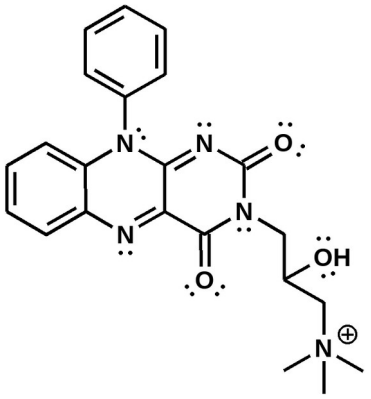
6. Functional groups and types of carbon. (a)–(c) Circle and name every functional group.



(a) acetaminophen

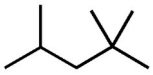


(b) aspirin

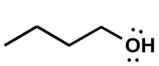


(c) Fl-3, a flavin from my Ph.D.

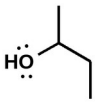
(d) Label every carbon in this molecule (2,2,4-trimethylpentane, the “isooctane” in gasoline octane ratings) as 1°, 2°, 3° or 4°.



(e)–(g) Is each alcohol primary, secondary or tertiary?



(e)



(f)



(g)

(e) _____ (f) _____ (g) _____

7. Degrees of unsaturation and isomers.

(a) Find the degrees of unsaturation: C_6H_6 ____ C_5H_8 ____ C_4H_7Br ____ C_3H_6O ____ C_3H_7NO ____

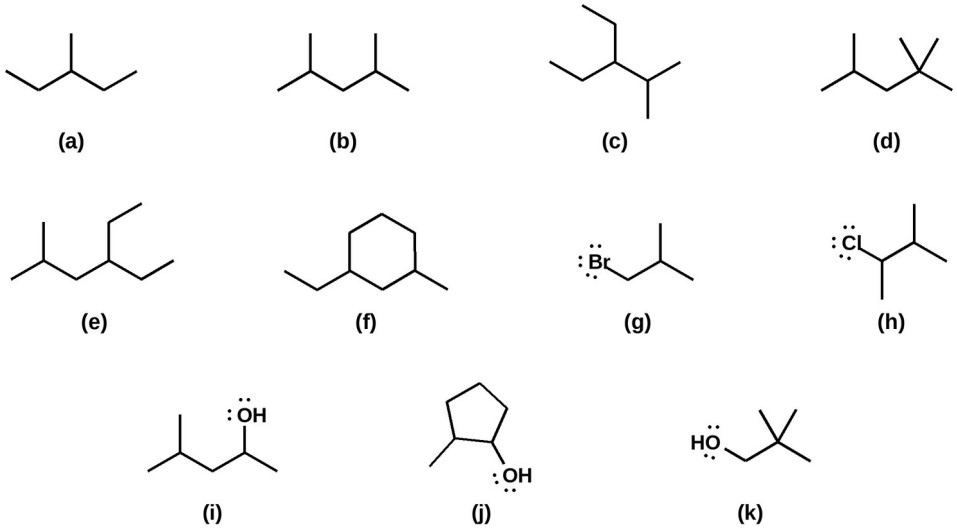
(b) For C_5H_8, give two different structures that fit: one with a ring and one without.

(c) Draw all the constitutional isomers of C_6H_{14}. How many are there?

(d) Draw all the constitutional isomers of C_3H_8O. Which one cannot form hydrogen bonds with itself?

(e) Draw all the constitutional isomers of C_4H_9Br and label each one 1°, 2° or 3°.

8. Name these compounds. Give the IUPAC name of each.



(a) _____ (b) _____

(c) _____ (d) _____

(e) _____ (f) _____

(g) _____ (h) _____

(i) _____ (j) _____

(k) _____

9. Draw the structure. Draw a bond-line structure for each name.

(a) 3-ethyl-2,2-dimethylpentane (b) 2-bromo-4-methylhexane (c) 1-chloro-3-ethylcyclopentane

(d) 2-methylbutan-2-ol (e) 3,3-dimethylcyclohexan-1-ol (f) 4-isopropylheptane

10. Intermolecular forces. Rank each set from lowest to highest boiling point and say which force makes the difference.

(a) butan-1-ol, butanal, pentane

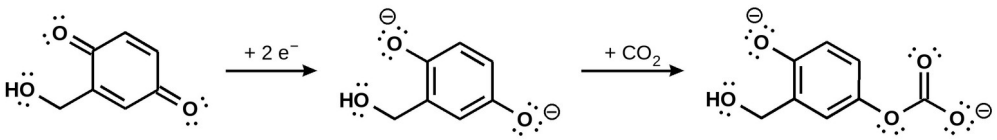
(b) hexane, 2,2-dimethylbutane, 2-methylpentane

(c) ethanol, propane, dimethyl ether (CH_3OCH_3)

(d) Which dissolves better in water, ethanol or hexan-1-ol? Why?

11. From my research: quinones and CO_2.

In my Ph.D. I made small quinones like HMQ A that take up two electrons at an electrode. Once reduced, the negatively charged oxygen can react with CO_2 and hold it as a carbonate. I added the CH_2OH arm so it could hydrogen-bond to the reduced quinone without blocking the spot where CO_2 attacks.



HMQ A dianion CO_2 adduct

(a) Give the molecular formula and the degrees of unsaturation of HMQ A. Name its functional groups.

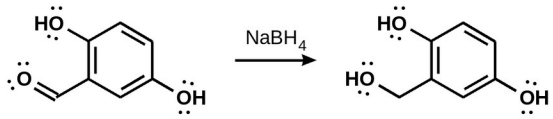
(b) In HMQ A, what is the hybridization of the ring carbons? Of the CH_2 carbon?

(c) The two O^- groups of the dianion push electrons into the ring. Draw resonance structures that move the negative charge onto ring carbons. Which ring carbons can carry the charge?

(d) The carbon of CO_2 is ____ hybridized with a ____° O–C–O angle. In the CO_2 adduct the same carbon is ____ hybridized with ____° angles.

(e) Draw the second resonance structure of the $–O–CO_2^-$ group in the adduct.

To make the hydroquinone HQ1, I reduced an aldehyde with sodium borohydride ($NaBH_4$):



2,5-dihydroxybenzaldehyde HQ1

(f) Which functional group changes? How does the hybridization of that carbon change?

(g) Give the molecular formula of each compound and the number of π bonds in each.

I also made flavins that dissolve in water, like Fl-3 in Problem 6. Fl-2 is the same molecule without the OH on the side chain. In water, Fl-3 dissolved to 80.3 mM and Fl-2 to 62.6 mM.

(h) What intermolecular force does the extra OH add? Use it to explain why Fl-3 dissolves better than Fl-2.

_____

_____